

Structured State Space Model Dynamics and Parametrization for Spiking Neural Networks

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Abstract

Multi-state spiking neurons such as the adaptive leaky integrate-and-fire (AdLIF) neuron offer compelling alternatives to conventional deep learning models thanks to their sparse binary activations, second-order nonlinear recurrent dynamics and efficient hardware realizations. However, such internal dynamics can cause instabilities during inference and training, often limiting performance and scalability. Meanwhile, state space models (SSMs) excel in long sequence processing using linear state-intrinsic recurrence resembling spiking neurons' subthreshold regime. Here, we establish a mathematical bridge between SSMs and second-order spiking neuron models. Based on structure and parametrization strategies of diagonal SSMs we propose two novel spiking neuron models. The first extends the AdLIF neuron through timestep training and logarithmic reparametrization to facilitate training and improve final performance. The second additionally brings initialization and structure from complex-state SSMs, broadening the dynamical regime to oscillatory dynamics. Together, our two models achieve beyond or near state-of-the-art (SOTA) performances for reset-based spiking neuron models across both event-based and raw audio speech recognition datasets. We achieve this with a favorable number of parameters and required dynamic memory while maintaining high activity sparsity. Our models demonstrate enhanced scalability in network size and strike a favorable balance between performance and efficiency with respect to SSM models. Our code is available at <https://github.com/Maxtimer97/SSM-inspired-LIF>.

1 Introduction

Spiking Neural Networks (SNNs) and their realization in neuromorphic hardware are gaining interest as promising low-power alternatives to artificial neural networks (ANNs) [1, 2]. On top of being compact recurrent units with strong temporal encoding capabilities, the biologically inspired binary spiking and the nonlinear feedback mechanism known as reset provide spiking neurons with highly efficient and sparse event-based communication. In the following, we refer to such models as reset-based spiking models, in contrast to non-reset models that only offer linear recurrence and non-spiking models that do not employ binary or at least graded spiking. Considering that sparse [3]

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and matmul-free [4] schemes have been receiving significant attention in more traditional models, like transformers, as a way to drastically reduce the computational complexity of ever-growing models, the avenue for high-performing, efficient SNNs is more relevant than ever.

Over the past decade, methods leveraging surrogate gradients to train SNNs similarly to recurrent neural networks (RNNs) using backpropagation through time (BPTT) have significantly advanced SNN performance on a variety of benchmarks [5, 6], as for instance the central adaptive leaky integrate-and-fire (AdLIF) model for audio classification tasks [7]. However, despite these successes, SNNs remain challenging to train and scale. They often exhibit unstable dynamics, which can hinder convergence and limit their applicability to larger, more complex tasks [8].

A potential avenue to addressing these challenges is through the perspective of state space models (SSMs), which have recently demonstrated superior performance in modeling long sequences, often outperforming traditional RNNs on various benchmarks [9–11]. This gain in performance on long sequences is not only due to the fact that SSMs constrain themselves to a linear recurrent state, unlike traditional RNNs or SNNs, but also to additional training optimizations. In fact, deep learning SSMs like the S4 model [9] incorporate architectural innovations, such as structured parameterizations and initialization, providing more robust and backpropagation-friendly recurrent state dynamics [12].

This study leverages the dynamic system modeling strengths of the S4 model to enhance nonlinear recurrent SNNs on spatiotemporal sequences. Specifically, we establish a parallel between the AdLIF model, the resonate-and-fire (RF) neuron, and SSMs. Building on these insights, we propose two models: (1) The SSM-inspired LIF (SiLIF) model, a S4-based enhancement of the AdLIF neuron with timestep training and logarithmic reparametrization. (2) the complex-valued SSM-inspired LIF (C-SiLIF) model, a novel top-down designed spiking neuron employing the diagonal S4 (S4D) [10] resonant dynamics, parametrization and initialization with a matching reset mechanism. Following previous work, we evaluate these reset-based spiking models on a suite of keyword classification datasets, both in the event-based and raw audio domains. Both SiLIF and C-SiLIF achieve state-of-the-art or highly competitive performance, offering improvements in training stability, accuracy and scalability due to the integration of SSM-inspired features. Our novel contributions are:

1. Establishing a theoretical bridge between SSMs and spiking neural networks (SNNs), revealing novel insights into their shared training dynamics.
2. Proposing two novel reset-based spiking neuron models that incorporate SSM-derived dynamics and parametrizations.
3. Demonstrating significant improvements in accuracy and scalability on audio classification tasks while maintaining efficiency, surpassing previous reset-based models.
4. Demonstrating Pareto optimal performance that balances accuracy and computational efficiency relative to SSMs for event-based data.

2 Related work

Spiking neural networks for spoken word classification tasks Training deep and high performance SNNs has become possible using techniques from deep learning, such as the surrogate gradient method [13], error backpropagation mechanisms [14], and ANN-to-SNN conversion [15].

One promising application for spiking neurons has been the classification of spoken words. This was demonstrated by Bittar and Garner [7], with the use of their specific adaptive leaky integrate-and-fire (AdLIF) neuron, inspired from previous neuroscience and neuromorphic studies [16–20], and which we re-introduce precisely in the Methods section. Different iterations refined the AdLIF model, either by constraining certain parameters [21], or by employing a different discretization approach [22], but they only led to marginal improvements on large scale datasets. Our work significantly advances this line by extending the AdLIF through SSM-inspired training dynamics.

Additionally, recent works have explored the application of the resonate-and-fire (RF) neuron model introduced by Izhikevich [23] in deep SNNs [24–26]. But despite their biological plausibility, RF-based SNNs have yet to achieve state-of-the-art (SOTA) performance within the deep learning paradigm. Very recently, Huber et al. [27] even proposed a scaled-up RF model employing SSM structure, but their performance still lacks behind previous spiking references. Our study identifies poor training initialization and parametrization as key bottlenecks in such models and proposes the first RF-style spiking neuron achieving SOTA performance on audio classification benchmarks.

Synaptic delays [28] enhance SNN models temporal processing capabilities, achieving SOTA accuracy on event-based datasets, while maintaining practical training times. Deckers et al. [21] employ these learnable delays on top of its constrained AdLIF model, giving an insight on the full range of

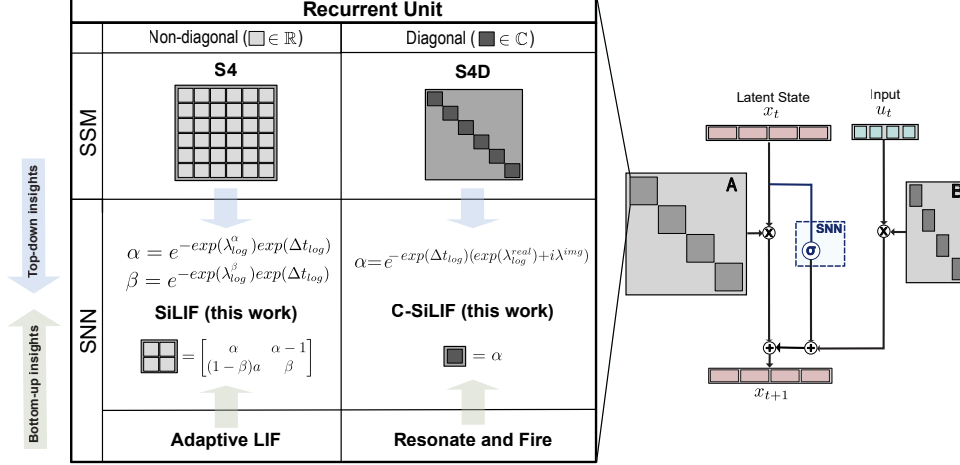


Figure 1: Proposed method to distill features of modern structured state space models (SSMs) to build novel high performing spiking neural networks (SNNs). The SiLIF model is an enhancement of the adaptive LIF (AdLIF) neuron through reparametrization of its state transition matrix following the S4 model. The C-SiLIF additionally exploits the complex representation and the specific initialization of the S4D model, drawing a parallel with the resonate-and-fire (RF) neuron. The right part of the figure illustrates the similar structure of SSMs and SNNs, plus the SNN-exclusive reset mechanism.

temporal processing capabilities of SNNs. But the implementation of synaptic delays on hardware requires a significant amount of dynamic memory, close to the static parameter memory, to store the delayed spikes. The feasibility of such an implementation and the effective advantage for neuromorphic solutions has yet to be proven. Our proposed models meet or exceed the performance of delay models without requiring additional on-chip memory.

State space models for spiking and event-driven deep neural networks State space models, which mostly consist of large expanded linear recurrent states (see more in Methods section), have emerged as a reference model for processing long sequences [9, 29, 30], pushing boundaries on the long range arena benchmark [31], where even transformer models were failing. Recent work explored integrating SSMs with spiking and event-driven systems to enhance their efficiency. Bal and Sengupta [32] append probabilistic binary spiking to a standard SSM to improve sparsity while maintaining strong sequential processing capabilities. Similarly, Huang et al. [33] introduced a parallel RF neuron, which models subthreshold neuronal oscillations in the complex domain. These works preserve binary spiking and enable fast parallel training by adopting the linear recurrent structure of SSMs. Deep SSMs without binary spiking were also applied to event-based sensory signals, leveraging their stability and parallelizability to process long event streams in full resolution [34]. In event-based vision, SSMs with learnable timescale parameters were employed to adapt to varying inference frequencies without retraining, improving generalization over different temporal resolutions [35].

Nevertheless, unlike reset-based spiking models, all the above studies rely on pure linear recurrent dynamics along with SSM state expansion. This high dimensional state structure is less suitable for efficient implementations on neuromorphic hardware, especially for processing shorter sequences like audio samples where their benefits are more marginal. In contrast, our models extract key insights from SSMs to reach new SOTA performances on keyword classification tasks while maintaining the high level of compactness and sparsity of SNN models.

3 Methods

State space models State space models are based on the linear latent state dynamics (Eq. 1 below). Each scalar input feature $u \in \mathbb{R}$ is expanded to N dimensions through a complex vector $\mathbf{B} \in \mathbb{C}^N$, yielding the complex vector $x \in \mathbb{C}^N$, that is then recursively updated through the transition matrix $\mathbf{A} \in \mathbb{C}^{N \times N}$. The readout $y \in \mathbb{R}$ is eventually defined as the real part of the state

projection through $\mathbf{C} \in \mathbb{C}^N$ plus the direct signal projection through $\mathbf{D} \in \mathbb{R}$. The discretized form (Eq. 2) for a timestep Δt is obtained with the zero-order hold (ZOH) method (Eq. 3).

$$\begin{aligned} \dot{x}(t) &= \mathbf{A}x(t) + \mathbf{B}u(t) & (1a) & & x_t &= \bar{\mathbf{A}}x_{t-1} + \bar{\mathbf{B}}u_t & (2a) \\ y(t) &= \text{Re}(\mathbf{C}x(t)) + \mathbf{D}u(t) & (1b) & & y_t &= \text{Re}(\bar{\mathbf{C}}x_t) + \bar{\mathbf{D}}u_t & (2b) \\ & \text{where } \bar{\mathbf{A}} = e^{\mathbf{A}\Delta t}, \quad \bar{\mathbf{B}} = \bar{\mathbf{A}}^{-1}(\bar{\mathbf{A}} - \mathbf{I})\mathbf{B}, \quad \bar{\mathbf{C}} = \mathbf{C}, \quad \bar{\mathbf{D}} = \mathbf{D}, & (3) \end{aligned}$$

where $\text{Re}(\cdot)$ refers to the real part. These dynamics generalize to an input of dimension H by instantiating separate SSM blocks for each input dimension resulting in block-diagonal matrices $\mathbf{A}$ and $\mathbf{B}$ as shown in Fig. 1. As proposed by Gu et al. [9], SSM blocks are then typically placed in successive layers interleaved with multi-layer perceptrons (MLP) and gated linear units (GLU).

AdLIF neurons in the SSM domain Spiking neurons with adaptive dynamics have been of long term interest to better match the behavior of biological neurons and allow for more intricate temporal processing [16–20]. Here we use the specific adaptive leaky integrative and fire model (AdLIF) which demonstrated first competitive performances on speech recognition tasks [7]. It extends the LIF model with an adaptation variable w in a feedback loop with the neuron’s membrane potential u , as follows:

$$\begin{aligned} u_t &= \alpha u_{t-1} + (1 - \alpha)(I_t - w_{t-1}) - \theta s_{t-1} \\ w_t &= \beta w_{t-1} + (1 - \beta)au_{t-1} + bs_{t-1} \\ s_t &= (u_t \geq \theta), \end{aligned} \quad (4)$$

where at timestep t , I_t is the input signal, s_t the potential binary spike and θ is the spiking threshold. Additionally, $\alpha = e^{-\Delta t/\tau_u}$ and $\beta = e^{-\Delta t/\tau_w}$ where Δt is the discrete timestep duration and τ_u and τ_w are the continuous time constants of the u and w variables. Following [7], the model is usually trained in the discrete domain, using the discrete decay constants α and β as training parameters without involving τ_u and τ_w . These are completed by the sub- and above-threshold feedback parameters a and b .

In its subthreshold regime, the AdLIF neuron (Eq. 4) can be mapped to a linear SSM block with two real-valued states using parameters

$$\bar{\mathbf{A}} = \begin{bmatrix} \alpha & \alpha - 1 \\ (1 - \beta)a & \beta \end{bmatrix}, \quad \bar{\mathbf{B}} = \begin{bmatrix} 1 - \alpha \\ 0 \end{bmatrix}, \quad \bar{\mathbf{C}} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \bar{\mathbf{D}} = 0. \quad (5)$$

In contrast to a SSM, a spike-triggered feedback capturing the neuron’s reset mechanism is applied to the states u and w through θ and b , respectively, making the state dynamics nonlinearly dependent on the neuron membrane potential. This biologically-inspired feature plays crucial role in sparsity and nonlinear adaptation in SNNs. With the perspective of building improved SNN systems, we thus present in the following sections two models that merge efficient SSM features with nonlinear adaptive spiking neuron models.

SiLIF Model: SSM-Inspired parametrization Here, we introduce our SSM-inspired LIF neuron (SiLIF) derived by progressively integrating SSM features into the AdLIF neuron. Firstly, we note that structured SSMs such as S4 [9] are generally used in their discretized form (Eq. 2) while employing the continuous domain state space variables $\mathbf{A}$ and $\mathbf{B}$ as trained parameters for gradient descent. Inspired by this approach, we avoid the direct training of the discrete decay factors α and β and instead optimize their continuous-domain counterparts $\lambda^\alpha = \frac{1}{\tau_u}$ and $\lambda^\beta = \frac{1}{\tau_w}$.

Unlike most prior SNN studies that treat the timestep Δt as a fixed dataset-dependent constant, following S4’s approach [9], we define it as a heterogeneous trainable parameter in the SiLIF. This additional flexibility allows the model to cover a wider range of transition dynamics regimes between its different neurons, as showed in Fig. 2. Additionally, we implement S4’s logarithmic reparameterization of the continuous parameters to enhance numerical stability and facilitate smoother optimization $\lambda^\alpha = \exp(\lambda_{log}^\alpha)$, $\lambda^\beta = \exp(\lambda_{log}^\beta)$, $\Delta t = \exp(\Delta t_{log})$. This transformation has been shown to reduce the risks of vanishing and exploding gradients, thus improving the training stability of recurrent architectures [12]. Additionally, since $\exp(\lambda_{log}^\alpha)$, $\exp(\lambda_{log}^\beta)$ and $\exp(\Delta t_{log})$ are strictly positive, this formulation ensures that the effective decay factors remain within a stable regime $0 \leq \alpha, \beta \leq 1$ without requiring explicit clamping, as previously employed in AdLIF models [7]. The resulting SiLIF model employing core AdLIF dynamics (Eq. 4) and SSM inspired enhanced parameterization with heterogeneously trained parameters is described in Alg. 1.

Algorithm 1: SiLIF: an SSM-inspired LIF neuron

Initialize trainable parameters:

$\lambda_{log}^\alpha \sim \mathcal{U}(\log(\lambda_{min}^\alpha), \log(\lambda_{max}^\alpha)), \lambda_{log}^\beta \sim \mathcal{U}(\log(\lambda_{min}^\beta), \log(\lambda_{max}^\beta)), \Delta t_{log} \leftarrow \log(\Delta t_0),$
 $a \sim \mathcal{U}(a_{min}, a_{max}), b \sim \mathcal{U}(b_{min}, b_{max})$

Initialize states:

$u \sim \mathcal{U}(0, 1), w \sim \mathcal{U}(0, 1), s \sim \mathcal{U}(0, 1)$

Forward pass:

$\alpha \leftarrow \exp(-\exp(\lambda_{log}^\alpha) \cdot \exp(\Delta t_{log})), \beta \leftarrow \exp(-\exp(\lambda_{log}^\beta) \cdot \exp(\Delta t_{log}))$

Clamp $a \in [a_{min}, a_{max}], b \in [b_{min}, b_{max}]$

for t in sequence **do**

$w \leftarrow \beta \cdot w + a \cdot u + b \cdot s$
 $u \leftarrow \alpha \cdot (u - s) + (1 - \alpha) \cdot (I_t - w)$
 $s \leftarrow (u \geq \theta)$

C-SiLIF model: SSM-inspired complex state State space models like S4 use complex-valued parameters and states, $x = x^{real} + ix^{img} \in \mathbb{C}$, which was recently proven to improve parameter efficiency and stability with respect to real parametrization [36]. A scalar complex transition $x_t = ax_{t-1} + bu_t$ with $a, b \in \mathbb{C}$, corresponds to the following 2x2 real-valued system:

$$\begin{bmatrix} x_t^{real} \\ x_t^{img} \end{bmatrix} = \begin{bmatrix} a^{real} & -a^{img} \\ a^{img} & a^{real} \end{bmatrix} \begin{bmatrix} x_{t-1}^{real} \\ x_{t-1}^{img} \end{bmatrix} + \begin{bmatrix} b^{real} \\ b^{img} \end{bmatrix} u_t. \quad (6)$$

This has a similar format to the state transition in the AdLIF and SiLIF models, except that a complex system exhibits a structured antisymmetric matrix. Such a 2x2 system encompasses all state dynamics present in the diagonalized, better performing, version of S4, the S4D model [37], as it employs a complex diagonal transition matrix $\tilde{\mathbf{A}}$. As demonstrated by Ran-Milo et al. [36], such complex parametrization allows for a more compact model with training-friendly moderate value parameters, which we corroborate later through reported performance on event-based datasets (Sec. 4).

Inspired by this complex constrained approach, we propose a second spiking model, the complex-valued SSM-inspired LIF (C-SiLIF). We employ a complex state and follow the same dynamics as the 2x2 system described in Eq. 6. While [10] typically freezes the input projection $b = 1 + i$ and optimize the output projection $\tilde{\mathbf{C}}$ during training, we empirically found that freezing $\tilde{\mathbf{C}} = \mathbf{1}$ and optimizing $b \in \mathbb{R}$ yields better performance for our model. This implies that the output is twice the real part of the state variable, which passes through a threshold to generate a spike. To complete our spiking neuron model, we incorporate a reset by introducing feedback from spiking events into the neuronal state. Since the output consists of twice the real part of the state variable, half the spike value is fed back into the state space to maintain consistent pre-synaptic dynamics. As for SiLIF model, we use heterogenous parameters, logarithmic reparametrization and trainable step size for the transition parameters α^{real} and α^{img} , to enhance the stability and capacity of the model. Additionally for the C-SiLIF, we empirically choose to utilize S4D-Lin initialization derived from orthogonal polynomial projections for optimal SSM dynamics [37].

We thus formulate the final version of our complex-valued diagonal SSM-inspired LIF (C-SiLIF) neuron as presented in Alg. 2. The proposed dynamics closely resemble the RF model [23], but the SSM-inspired parameterization, initialization, and discretization enhance the stability [12], dynamical richness (Fig. 2) and performance (Sec. 4) of our model drastically compared to RF models.

Algorithm 2: C-SiLIF: an SSM-inspired LIF with complex state

Initialize trainable parameters:

$\lambda_{log}^{real} \leftarrow \log(0.5), \lambda_{log}^{img} \leftarrow \pi, \Delta t_{log} \sim \mathcal{U}(\log(\Delta t_{min}), \log(\Delta t_{max})), b \sim \mathcal{U}(0, 1),$

Initialize states:

$u \sim \mathcal{U}(0, 1), s \sim \mathcal{U}(0, 1)$

Forward pass:

$\alpha \leftarrow \exp((- \exp(\lambda_{log}^{real}) + i \cdot \lambda_{log}^{img}) \cdot \exp(\Delta t_{log}))$

for t in sequence **do**

$u \leftarrow \alpha \cdot (u - 0.5 \cdot s) + b \cdot I_t$
 $s \leftarrow (2\text{Re}(u) \geq \theta)$

Neuronal dynamics regimes We examine the impact of the SSM-inspired features introduced on the dynamics of our C-SiLIF and SiLIF models by representing the distribution of eigenvalues of the transition matrix $\tilde{\mathbf{A}}$ in the complex plane and comparing these to the RF model and a constrained version of the AdLIF, the cAdLIF [21]. The cAdLIF simply constrains the adaptive parameter a in Eq. 4 to positive values, which helps stability and yields improved performances. The distributions are obtained after training models on the SSC dataset (see Sec. 4) and presented in Fig. 2.

The cAdLIF model displays the most constrained eigenvalues, with mostly real values above 0.5. In fact, cAdLIF and AdLIF use strict constraints on its parameters range, with decay parameters α and β below but close to 1 and the adaptation parameter a within $[1, -1]$, which mathematically leads to mostly real eigenvalues. The RF model with its complex-constrained 2x2 system and fixed timestep Δt leads to a very concentrated eigenvalue distribution around unity with limited imaginary components.

On the other hand, our SiLIF and C-SiLIF models cover a way bigger range of neuronal dynamics. The SiLIF covers most of the real-positive unit circle due to the absence of clamping on the transition parameters α and β and the Δt training which can lead to very small transition values. Additionally the complex system defined by the C-SiLIF allows for transition parameters to be negative, thus leading to an eigenvalue distribution covering the whole unit circle. This allows our models to produce a wider, more flexible range of trained neuronal dynamics, including oscillatory ones which the cAdLIF or AdLIF typically cannot, as shown in the Appendix.

We demonstrate in the Experiments section below that these widened neuronal dynamics arising from Δt 's training, the absence of parameter clamping or the specific complex initialization significantly improve performance on audio classification tasks, establishing our SSM-inspired models as preferable choices.

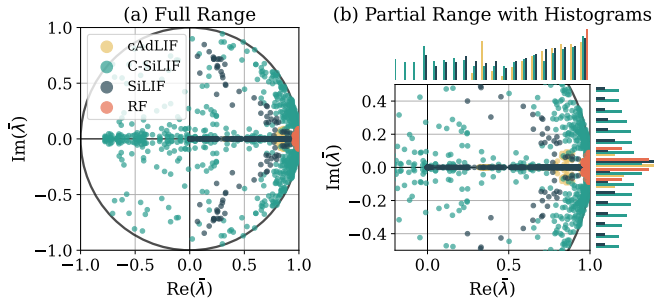


Figure 2: Scatter plot and histogram of state transition matrix eigenvalues for different pre-trained neuron models. (a) depicts the full unit circle while (b) shows a zoomed view with a logarithmic range histogram of the eigenvalues. The range of covered eigenvalues, especially out of the real axis, correspond to different accessible neuronal dynamics regimes.

4 Experiments

Experimental setup We evaluate our model on keyword classification tasks, which are common benchmarks for evaluating spiking and recurrent neural networks. We namely use the Spiking Heidelberg Digits (SHD), the Spiking Speech Commands (SSC) [38], and the Google Speech Commands v0.02 (GSC) dataset [39], in accordance to recent research on effective applicative neuromorphic models [7, 21, 22, 27, 28, 34]. The first two datasets consist of a stream of binary events data, obtained from the original audio signal using the cochlea model proposed by Cramer et. al. [38]. The SHD dataset consists of 10k audio samples equally distributed over 20 classes of German and English spoken digits. GSC presents a larger dataset with more than 100k samples distributed over 35 classes of spoken command words, identically to SSC which is GSC's event-based converted version.

Different studies have used varying temporal binning and resolution for the spiking datasets, which has a significant impact on classification performance. Thus we handle multiple cases to present fair comparisons: by default, we employ bins of 4ms, and scale down to 10ms or 14ms bins for comparison when required. Additionally the spiking datasets are binned spatially from 700 to 140 input channels matching recent studies. Considering GSC, the samples are fixed to 10ms time bins. They are additionally passed through a Mel filterbank with 40 Mel filters, matching the setup used in all similar studies. Finally, we note that the SHD dataset doesn't contain a separate test and validation set, which led previous studies to report the model's validation accuracy at the best epoch. We follow the same approach here for fair comparison.

Our implementation is based on the SpArch² code by Bittar and Garner [7], which we build upon. The corresponding network architecture has since been employed by most works in this field [21, 27, 28]. The network consists of two non-recurrent hidden layers with batch normalization and dropout, and the output layer is fixed to a non-firing leaky integrate (LI) neuron whose outputs are passed through a softmax and summed over time. For our SiLIF and C-SiLIF models, the trainable neuron parameters and their initializations are to be found in Algorithms 1 and 2, respectively. The spiking threshold is fixed at 1. We use a cross-entropy loss with an Adam optimizer and a Plateau scheduler. We selected hyperparameters for our models based on grid searches of the learning rate, dropout and scheduler (Table 1). Other parameters like precise initialization ranges can be found directly in our code. All of our reported results are averaged results over 5 and 10 different seeds runs respectively for SHD, and SSC and GSC. The average μ and the standard deviation σ are reported as $\mu \pm \sigma\%$.

Altogether this project required about 4000 GPU hours on a Nvidia RTX 4090 with 24 GB memory with shared 377 GB AMD EPYC 9384X 32-Core CPU on an internal cluster, also considering initial exploratory runs. Each run takes in average 2 hours for the SHD dataset, 15 hours for SSC and 5 hours for GSC. This allowed for detailed ablation study and reproduction of previous results with grid search analysis, targeting the most detailed and representative results possible.

Table 1: Experimental setups for our models on different datasets.

Dataset	Model	Resolution	#Hidden Size	lr	Dropout	Scheduler (Patience/Factor)
SHD	C-SiLIF	4 ms	128	0.01	0.5	5/0.9
		4 ms	512	0.005	0.1	10/0.9
SSC	C-SiLIF	10 ms/4 ms	512	0.005/0.01	0.5	10/0.7
	SiLIF	10 ms/4 ms	512	0.01	0.5	10/0.7
GSC	C-SiLIF	10 ms	512	0.001	0.25	5/0.7
	SiLIF	10 ms	512/1024	0.001	0.5	5/0.7

Results on event-based keyword classification tasks We present results of our two models C-SiLIF and SiLIF on the SHD, SSC and GSC datasets in Table 2 and compare these to the previous state-of the art obtained with reset-based spiking neuron models. We test different temporal resolutions, report the total number of parameters and an estimation of the required number of dynamic state buffers for each model. This last metric highlights the dynamic memory cost of each model, which can be large *e.g.* when using synaptic delays [21, 28]. In fact, in a streamlined processing of data, non-delay models only require their neuron states to be updated and maintained in a real value dynamic memory, or buffer, at each timestep. On the other hand, delay-based models require extra memory (buffers) to store delayed inputs per synapse before they are summed and sent to the corresponding neuron. For a detailed justification of the buffer values reported for the different models, refer to the Appendix. We also note that the event-by-event SSM model from Schöne et al. [34] is not included in this table as it doesn’t use reset-based neurons. We report more on efficiency and compare this solution to reset-based models in the next paragraph (Fig. 3).

On the SHD task, our C-SiLIF model outperforms previous state-of-the-art accuracy for networks below 50k parameters with 95.06% accuracy. At higher scale, it nears the current state-of-art obtained for the Simplectic-Euler (SE)-adLIF model [22], within its margin of error. On the more representative SSC and GSC tasks, our SiLIF model significantly outperforms all previous non delay-based state-of-the-art accuracy. For SSC, both our models even outperform delay-based models with significant gains, making efficient use of the high 4 ms resolution, which delay-based models would pay a high buffer cost to run. At 10 ms resolution, the SiLIF is on par with the synaptic delay cAdLIF (d-cAdLIF) [21] and the delay-LIF model DCLS-Delays [28] with half the parameters and significantly less required buffers. On GSC, our 512 units SiLIF model significantly outperforms the cAdLIF accuracy, reaching the best performance for models below 0.3M parameters. Our scaled 1024 units SiLIF also reaches the second best performance among all models, just below the d-cAdLIF model, with twice the number of parameters but 75 times less required buffers.

²<https://github.com/idiap/sparch>, released under BSD-3-Clause License.

Table 2: Classification accuracy on SHD, SSC and GSC datasets in comparison with other state of the art reset-based spiking models. For each task, models are arranged per number of parameters and the best model is in bold while the second best is underlined.

Dataset	Method	Resolution	# Params	# Buffers	Top1 Accuracy
SHD	SE-adLIF, 1 layer [22]	4 ms	37.5k	0.3k	94.59 $\pm$ 0.27%
	cAdLIF [21]	10 ms	38.7k	0.5k	94.19%
	C-SiLIF, # hidden 128 (ours)	4 ms	38.7k	0.5k	95.06 $\pm$ 0.37%
	d-cAdLIF [21]	10 ms	76k	38.7k	94.85 $\pm$ 0.64%
	S5-RF [27]	4 ms	0.2M	1k	91.86 %
	DCLS-Delays [28]	10 ms	0.2M	0.1M	95.07 $\pm$ 0.24%
	C-SiLIF, # hidden 512 (ours)	<u>4 ms</u>	<u>0.35M</u>	<u>2k</u>	<u>95.55 $\pm$ 0.28%</u>
	SE-adLIF, 2 layer [22]	4 ms	0.45M	1.4k	95.81 $\pm$ 0.56%
	RadLIF [7]	14 ms	3.9M	4k	94.62%
SSC	cAdLIF [21]	10 ms	0.35M	2k	77.50%
	SiLIF, 10 ms bins (ours)	10 ms	0.35M	2k	80.11 $\pm$ 0.31%
	C-SiLIF, 4 ms bins (ours)	<u>4 ms</u>	<u>0.35M</u>	<u>2k</u>	<u>81.59 $\pm$ 0.31%</u>
	SiLIF, 4 ms bins (ours)	4 ms	0.35M	2k	82.03 $\pm$ 0.25%
	d-cAdLIF [21]	10 ms	0.7M	0.35M	80.23 $\pm$ 0.07%
	DCLS-Delays [28]	10 ms	1.2M	0.6M	80.29 $\pm$ 0.06%
	SE-adLIF [22]	4 ms	1.6M	2.9k	80.44 $\pm$ 0.26%
	S5-RF [27]	4 ms	1.7M	6k	78.8 %
	RadLIF [7]	14 ms	3.9M	4k	77.40%
GSC	cAdLIF [21]	10 ms	0.3M	2k	94.67%
	SiLIF, # hidden 512 (ours)	10 ms	0.3M	2k	95.25 $\pm$ 0.12%
	d-cAdLIF [21]	10 ms	0.61M	0.3M	95.69 $\pm$ 0.03%
	RadLIF [7]	10 ms	0.83M	2k	94.51%
	SiLIF, # hidden 1024 (ours)	<u>10 ms</u>	<u>1.1M</u>	<u>4k</u>	<u>95.49 $\pm$ 0.09%</u>
	DCLS-Delays [28]	10 ms	1.2M	0.6M	95.29 $\pm$ 0.11%

Computational cost vs accuracy analysis One of the main motivations to use spiking neurons is its potential for reduced energy consumption on dedicated event-based hardware, such as the Intel Loihi chip [40]. For this, we study models’ efficiency by reporting the synaptic operations (SOP) per sample. The SOP is the number of times a non-zero activity passes through a synaptic weight. In SNNs, this number is highly reduced as SOPs are only required when neurons produce a spike. We report accuracy and SOP from experiments on the non-delay AdLIF [7], cAdLIF [21] and our C-SiLIF and SiLIF models with all architectures consisting of 2 hidden layers of 512 neurons and training without any regularization loss or other sparsity enhancing method. Each experiment is run with 5 random seeds, defining hyperparameters with a grid search for the 4 ms resolution version of SHD and SSC and employing established values from corresponding studies on the fixed 10 ms GSC.

We additionally compare these models to reported results from the S5-RF [27] and Event-SSM [34] works. The S5-RF is another SSM-inspired spiking neuron model employing the HIPPO initialization but no further reparametrization, which is a key feature of our work. Their reported SOP are per timestep so we scale them by the number of timesteps per sample to match our metric. The Event-SSM is a linear model made to process event streams but without any spiking or reset mechanism, i.e. without any sparsity. Their SOP value is calculated analytically as detailed in the Appendix.

The results are reported in Fig. 3. We do not report the AdLIF on SSC as the model didn’t converge in our experiments and the S5-RF and Event-SSM for GSC as these models are only run on event-based tasks.

First, since the SHD and SSC experiments here are conducted at a 4 ms resolution, we confirm that our C-SiLIF and SiLIF models, respectively, outperform all reset-based models reproduced. Our C-SiLIF and SiLIF models also achieve a Pareto optimal among other reported models for accuracy and SOP on the SHD and SSC and GSC tasks respectively. Despite achieving top accuracy on the SSC task, the Event-SSM requires close to 10 times more operations than SSN models, in addition to costly real-valued activations instead of binary spikes. This confirms the favorable choice of our reset-based spiking models for efficient implementation on neuromorphic hardware.

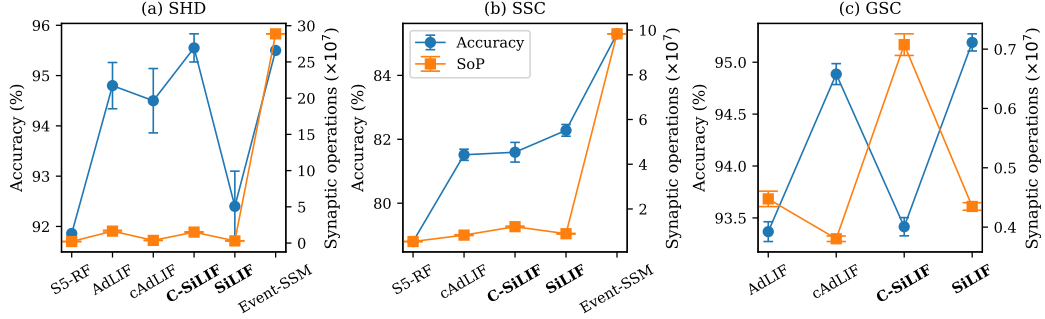


Figure 3: Test accuracy and synaptic operations (SOP) with standard deviation for different models on the 3 audio datasets SHD, SSC and GSC from left to right. Our models are shown in bold.

Ablation study We eventually conduct an ablation study of the C-SiLIF model, considering it includes all SiLIF features plus the complex-value constraint and S4D initialization (Table 3). The ablation mostly highlights the importance of the logarithmic reparametrization and the timestep Δt , which are in fact the features common to our two models. The first one confirms the positive impact of such a reparametrization for training spiking neurons. Additionally, we show for the first time the importance of an heterogeneous, or even trained Δt for reset-based spiking neurons. As seen in Fig. 2, this directly impacts the range of transition parameters and thus dynamic regimes covered by the model, which we now show to correlate with its performance. We also show that these features grant the C-SiLIF with favorable scalability in network size with respect to other reset-based models in the Appendix.

Table 3: Test accuracy on the SHD dataset with ablation of SSM-inspired features of the C-SiLIF model. The configurations represent the best accuracy hyperparameters obtained from a 5 seed sweep for each model with 2 hidden layers of 512 neurons.

Model	Accuracy	Configuration
C-SiLIF	$95.6 \pm 0.3\%$	lr=0.005, dropout=0.1
C-SiLIF w/o half reset and input gating	$94.9 \pm 0.3\%$	lr=0.005, dropout=0.1
C-SiLIF w/o half reset, input gating and S4D-Lin init.	$93.3 \pm 0.5\%$	lr=0.001, dropout=0.5
C-SiLIF w/o half reset, input gating, S4D-Lin init. and logarithmic reparametrization	$90.2 \pm 0.6\%$	lr=0.001, dropout=0.1
C-SiLIF with Δt constant (uniform distribution)	$95.1 \pm 0.7\%$	lr=0.01, dropout=0.5
C-SiLIF with Δt constant (single value)	$89.3 \pm 1.2\%$	lr=0.005, dropout=0.1

5 Conclusion

This work presents a theoretical bridge between state space models (SSMs) and spiking neurons and proposes two novel spiking neuron models integrating dynamics and parametrization inspired by SSMs. Especially, we show the importance of heterogeneous timestep parameters Δt along with the use of logarithmic reparametrization, which lead to a wider range of dynamical regimes, and eventually to improved performances. Our models outperform previous reset-based spiking neuron architectures on the event-based SSC dataset, deliver state-of-the-art (SOTA) performance for small models on the SHD task, and achieve near SNN SOTA accuracy—surpassed only by a delay-based model demanding 70 times more dynamic buffering—on GSC, while demonstrating enhanced capacity to scale with network size. Our models prevalence extends to non-reset models, where they are Pareto optimal in performance-efficiency with respect to SSM-SNN hybrids. Future work could investigate the use of the SSM state-expansion in spiking neurons, or applying our method to other network architectures or sensory processing tasks.

6 Limitations

The primary limitation of our paper lies in the employed datasets which were required for comparison to previous works. In fact the cochlea event-based conversion for the SHD and SSC datasets doesn't correspond to a real-world sensor, but our results on the large scale raw audio GSC dataset consolidates our evaluation. Additionally one could regret the lack of a "SiLIF + delay" model in the vein of the d-cAdLIF [21]. However, considering the implementation concerns for delay-based models and due to the substantial additional efforts required, this is beyond the scope of this work.

Acknowledgments

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References

- [1] Jason Yik, Korneel Van den Berghe, Douwe den Blanken, Younes Bouhadjar, Maxime Fabre, Paul Hueber, Weijie Ke, Mina A Khoei, Denis Kleyko, Noah Pacik-Nelson, et al. The neurobench framework for benchmarking neuromorphic computing algorithms and systems. *Nature Communications*, 16(1):1545, 2025.
- [2] Caterina Caccavella, Federico Paredes-Vallés, Marco Cannici, and Lyes Khacef. Low-power event-based face detection with asynchronous neuromorphic hardware. In *2024 International Joint Conference on Neural Networks (IJCNN)*, pages 1–10, 2024. doi: 10.1109/IJCNN60899.2024.10650843.
- [3] Iman Mirzadeh, Keivan Alizadeh, Sachin Mehta, Carlo C Del Mundo, Oncel Tuzel, Golnoosh Samei, Mohammad Rastegari, and Mehrdad Farajtabar. Relu strikes back: Exploiting activation sparsity in large language models, 2023. URL <https://arxiv.org/abs/2310.04564>.
- [4] Shuming Ma, Hongyu Wang, Lingxiao Ma, Lei Wang, Wenhui Wang, Shaohan Huang, Li Dong, Ruiping Wang, Jilong Xue, and Furu Wei. The era of 1-bit llms: All large language models are in 1.58 bits, 2024. URL <https://arxiv.org/abs/2402.17764>.
- [5] Jacques Kaiser, Hesham Mostafa, and Emre Neftci. Synaptic plasticity dynamics for deep continuous local learning (decolle). *Frontiers in Neuroscience*, 14:424, 2020. ISSN 1662453X. doi: 10.3389/fnins.2020.00424. URL <https://www.frontiersin.org/article/10.3389/fnins.2020.00424>.
- [6] Friedemann Zenke and Surya Ganguli. Superspike: Supervised learning in multilayer spiking neural networks. *Neural computation*, 30(6):15141541, 2018.
- [7] Alexandre Bittar and Philip N. Garner. A surrogate gradient spiking baseline for speech command recognition. *Frontiers in Neuroscience*, 16, 2022. ISSN 1662453X. doi: 10.3389/fnins.2022.865897.
- [8] Luca Herranz-Celotti and Jean Rouat. Stabilizing spiking neuron training. *arXiv preprint arXiv:2202.00282*, 2022.
- [9] Albert Gu, Karan Goel, and Christopher Ré. Efficiently modeling long sequences with structured state spaces. In *ICLR 2022 - 10th International Conference on Learning Representations*, 2022.
- [10] Ankit Gupta, Albert Gu, and Jonathan Berant. Diagonal state spaces are as effective as structured state spaces. In *Advances in Neural Information Processing Systems*, volume 35, 2022.
- [11] Antonio Orvieto, Samuel L Smith, Albert Gu, Anushan Fernando, Caglar Gulcehre, Razvan Pascanu, and Soham De. Resurrecting Recurrent Neural Networks for Long Sequences. pages 1–30, 2023. URL <http://arxiv.org/abs/2303.06349>.

- [12] Nicolas Zucchet and Antonio Orvieto. Recurrent neural networks: vanishing and exploding gradients are not the end of the story, 2024. URL <https://arxiv.org/abs/2405.21064>.
- [13] Emre O. Neftci, Hesham Mostafa, and Friedemann Zenke. Surrogate gradient learning in spiking neural networks: Bringing the power of gradient-based optimization to spiking neural networks. *IEEE Signal Processing Magazine*, 36(6):51–63, 2019. doi: 10.1109/MSP.2019.2931595.
- [14] Jun Haeng Lee, Tobi Delbruck, and Michael Pfeiffer. Training deep spiking neural networks using backpropagation. *Frontiers in Neuroscience*, 10, 2016. ISSN 1662453X. doi: 10.3389/fnins.2016.00508.
- [15] Tong Bu, Wei Fang, Jianhao Ding, Peng Lin Dai, Zhaofei Yu, and Tiejun Huang. Optimal ann-snn conversion for high-accuracy and ultra-low-latency spiking neural networks. In *ICLR 2022 - 10th International Conference on Learning Representations*, 2022.
- [16] E.M. Izhikevich. Simple model of spiking neurons. *IEEE Transactions on Neural Networks*, 14(6):15691572, 2003.
- [17] Romain Brette and Wulfram Gerstner. Adaptive exponential integrate-and-fire model as an effective description of neuronal activity. *Journal of Neurophysiology*, 94, 2005. ISSN 00223077. doi: 10.1152/jn.00686.2005.
- [18] Wulfram Gerstner, Werner M Kistler, Richard Naud, and Liam Paninski. *Neuronal dynamics: From single neurons to networks and models of cognition*. Cambridge University Press, 2014.
- [19] Darjan Salaj, Anand Subramoney, Ceca Krausnikovic, Guillaume Bellec, Robert Legenstein, and Wolfgang Maass. Spike frequency adaptation supports network computations on temporally dispersed information. *eLife*, 10, 2021. ISSN 2050084X. doi: 10.7554/eLife.65459.
- [20] Bojian Yin, Federico Corradi, and Sander M. Bohté. Accurate and efficient time-domain classification with adaptive spiking recurrent neural networks. *Nature Machine Intelligence*, 3, 2021. ISSN 25225839. doi: 10.1038/s42256-021-00397-w.
- [21] Lucas Deckers, Laurens Van Damme, Ing Jyh Tsang, Werner Van Leekwijck, and Steven Latré. Co-learning synaptic delays, weights and adaptation in spiking neural networks, 2023. URL <https://arxiv.org/abs/2311.16112>.
- [22] Maximilian Baronig, Romain Ferrand, Silvester Sabathiel, and Robert Legenstein. Advancing spatiotemporal processing in spiking neural networks through adaptation, 2024. URL <https://arxiv.org/abs/2408.07517>.
- [23] Eugene M. Izhikevich. Resonate-and-fire neurons. *Neural Networks*, 14, 2001. ISSN 08936080. doi: 10.1016/S0893-6080(01)00078-8.
- [24] Badr AlKhamissi, Muhammad ElNokrashy, and David Bernal-Casas. Deep spiking neural networks with resonate-and-fire neurons, 2021. URL <https://arxiv.org/abs/2109.08234>.
- [25] E. Paxon Frady, Sophia Sanborn, Sumit Bam Shrestha, Daniel Ben Dayan Rubin, Garrick Orchard, Friedrich T. Sommer, and Mike Davies. Efficient neuromorphic signal processing with resonator neurons. *Journal of Signal Processing Systems*, 94, 2022. ISSN 19398115. doi: 10.1007/s11265-022-01772-5.
- [26] Saya Higuchi, Sebastian Kairat, Sander M. Bohte, and Sebastian Otte. Balanced resonate-and-fire neurons, 2024. URL <https://arxiv.org/abs/2402.14603>.
- [27] Thomas E Huber, Jules Lecomte, Borislav Polovnikov, and Axel von Arnim. Scaling up resonate-and-fire networks for fast deep learning. *arXiv preprint arXiv:2504.00719*, 2025.
- [28] Ilyass Hammouamri, Ismail Khalfaoui-Hassani, and Timothée Masquelier. Learning delays in spiking neural networks using dilated convolutions with learnable spacings, 2023. URL <https://arxiv.org/abs/2306.17670>.
- [29] Albert Gu, Tri Dao, Stefano Ermon, Atri Rudra, and Christopher Re. Hippo: Recurrent memory with optimal polynomial projections, 2020. URL <https://arxiv.org/abs/2008.07669>.

- [30] Jimmy T. H. Smith, Andrew Warrington, and Scott W. Linderman. Simplified state space layers for sequence modeling, 2023. URL <https://arxiv.org/abs/2208.04933>.
- [31] Yi Tay, Mostafa Dehghani, Samira Abnar, Yikang Shen, Dara Bahri, Philip Pham, Jinfeng Rao, Liu Yang, Sebastian Ruder, and Donald Metzler. Long range arena: A benchmark for efficient transformers. *arXiv preprint arXiv:2011.04006*, 2020.
- [32] Malyaban Bal and Abhronil Sengupta. P-spikessm: Harnessing probabilistic spiking state space models for long-range dependency tasks, 2024. URL <https://arxiv.org/abs/2406.02923>.
- [33] Yulong Huang, Zunchang Liu, Changchun Feng, Xiaopeng Lin, Hongwei Ren, Haotian Fu, Yue Zhou, Hong Xing, and Bojun Cheng. Prf: Parallel resonate and fire neuron for long sequence learning in spiking neural networks, 2024. URL <https://arxiv.org/abs/2410.03530>.
- [34] Mark Schöne, Neeraj Mohan Sushma, Jingyue Zhuge, Christian Mayr, Anand Subramoney, and David Kappel. Scalable event-by-event processing of neuromorphic sensory signals with deep state-space models, 2024. URL <https://arxiv.org/abs/2404.18508>.
- [35] Nikola Zubić, Mathias Gehrig, and Davide Scaramuzza. State space models for event cameras, 2024. URL <https://arxiv.org/abs/2402.15584>.
- [36] Yuval Ran-Milo, Eden Lumbroso, Edo Cohen-Karlik, Raja Giryes, Amir Globerson, and Nadav Cohen. Provable benefits of complex parameterizations for structured state space models. *Advances in Neural Information Processing Systems*, 37:115906–115939, 2024.
- [37] Albert Gu, Ankit Gupta, Karan Goel, and Christopher Ré. On the parameterization and initialization of diagonal state space models. In *Advances in Neural Information Processing Systems*, volume 35, 2022.
- [38] Benjamin Cramer, Yannik Stradmann, Johannes Schemmel, and Friedemann Zenke. The heidelberg spiking data sets for the systematic evaluation of spiking neural networks. *IEEE Transactions on Neural Networks and Learning Systems*, 2020.
- [39] Pete Warden. Speech commands: A dataset for limitedvocabulary speech recognition. *arXiv preprint arXiv:1804.03209*, 2018.
- [40] M. Davies, N. Srinivasa, T. H. Lin, G. Chinya, P. Joshi, A. Lines, A. Wild, and H. Wang. Loihi: A neuromorphic manycore processor with onchip learning. *IEEE Micro*, PP(99):11, 2018. ISSN 02721732. doi: 10.1109/MM.2018.112130359.

A Technical Appendices and Supplementary Material

A.1 Link between eigenvalues and neuronal dynamics regimes

We give extra insights on the importance of the transition matrix eigenvalues of spiking neurons and their link to the dynamical regimes of the system.

The behavior of neurons with second order dynamics, including cAdLIF and SiLIF, can be classified as resonator, integrator, or mixed, depending on the ratio of the membrane and adaptation time constants ($\tau_m/\tau_w = \beta/\alpha$) and the coupling constant (a) (see Equation 4). These parameters determine whether the system undergoes a Hopf or saddle-node bifurcation which is why during training they are typically clamped within specific ranges to control neuronal behavior [7]. As seen in Figure 4, cAdLIF and SiLIF models exhibit both integrator and resonator regimes. The resonator regime corresponds to complex eigenvalues with a non-zero imaginary part, while the integrator regime is associated with purely positive real eigenvalues, as illustrated in Figure 2. Since α and β are not explicitly clamped for the SiLIF model, oscillatory dynamics are more thoroughly explored by the neuron, as there is no artificial constraint on the β/α ratio. This leads to more parameter exploration flexibility during training and richer internal dynamics. Nevertheless, due to the exponential reparameterization, the SiLIF model always remains in the stable regime.

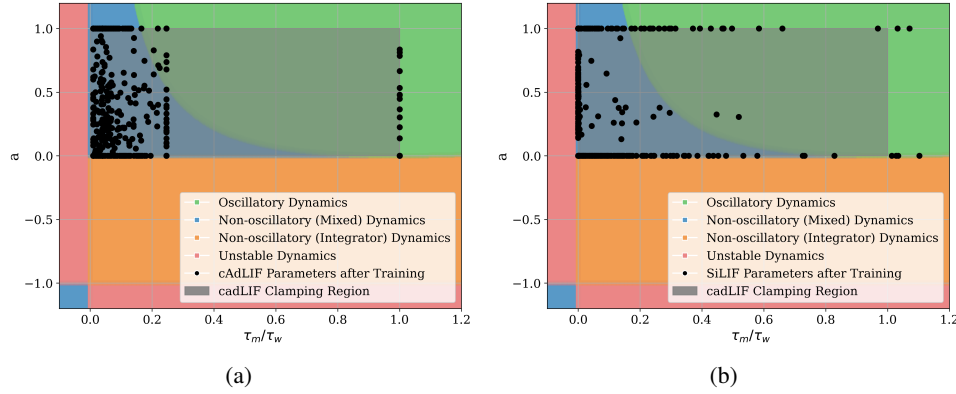


Figure 4: Neuronal dynamics of the (a) CadLIF and (b) SiLIF models pre-trained on the SSC task.

A.2 Hyperparameters search

We detail here the hyperparameters that are searched for each of the three audio datasets. These grid searches are performed both for our C-SiLIF and SiLIF models and for all reproduced results on the AdLIF, cAdLIF and RF models to ensure the most robust results in terms of training optimization. The only exception is for the AdLIF and cAdLIF results on GSC for Figure 3 where configurations from corresponding papers are kept considering that the dataset and resolution are exactly the same.

Table 4: Hyperparameter grid search space per dataset.

Hyperparameter	SHD	SSC	GSC
lr	{0.001, 0.005, 0.01}	{0.001, 0.005, 0.01}	{0.001, 0.005, 0.01}
dropout	{0.1, 0.25, 0.5}	{0.1, 0.25, 0.5}	{0.1, 0.25, 0.5}
schedule_patience	{5, 10*}	{5, 10}	{5, 10}
schedule_factor	{0.7, 0.9*}	0.7	0.7
C-SiLIF $\Delta t_{\max}$	0.5	5	5
batch size	128	256	32
epochs	100	100	100

* Fixed for ablation study (Table 3).

A.3 Buffers Computation

We give more details on the calculation of dynamic memory for SNN models in terms of number of buffers as employed in Table 2. This value provides a conservative lower-bound estimate, assuming sequential timestep processing, and serves as an indicator of practical hardware resource usage (footprint, energy, latency) for SNN implementations.

Non-delay models only require dynamic memory to store the state of each of their neurons, as the rest of the network’s activity can be flushed from one timestep to the next. Thus, the number of buffers $N_{buffers}$ is equal to $\sum_{layers} N_{hiddeens} \times N_{states}$ where $N_{hiddeens}$ is the number of hidden neurons per layer and N_{states} the number of real-valued states per neuron on this layer. Considering all considered non-delay models use two neuron states, their number of buffers is simply $\sum_{layers} 2N_{hiddeens}$.

For delay models, as such a metric hasn’t been considered or reported before, we propose a very conservative and favorable method to determine it. We first note that our two studied delay-based the d-cAdLIF [21] and the DCLS-Delays [28], exhibit a sparsity level around 95%, meaning that each neuron produces on average a spike every 20 timesteps. Considering that the maximum delay applied to each spike is of 25 steps for the d-cAdLIF and 30 for the DCLS-Delays, in an optimistic scenario assuming uniform spiking activity, around 1 buffer will be required to maintain a delayed spike per synapse at every point in time. This omits the case of a burst of spikes in multiple channels, which would cause an increase in required buffers to cover this punctual event and thus the minimum number of required buffer. Thus for these models, we set $N_{buffers} = \sum_{layers} N_{hiddeens} \times N_{states} + N_{synapses}$ with $N_{synapses}$ the total number of synapses of the model. This corresponds to a significant increase in number of buffers compared to non-delay models as the number of synapses scales quadratically with the number of hidden units, as reported in Table 2.

A.4 Event-SSM synaptic operations computation

We detail here the computation to determine the number of synaptic operations (SOPs) required per sample for the Event-SSM model [34]. This model achieves state-of-the-art performance on the SSC dataset but relies on projecting each event to a real-value vector and processing them without any sparsity, thus requiring a significant number of SOPs. We go here through the SOPs computation for the smaller Event-SSM on SSC with 64 hidden states which is reported in Figure 3.

First, the Event-SSM fully relies on event-based processing and doesn’t contain any sparsity mechanism meaning each event triggers the same number of fixed operations through the system. The first operation each event goes through is an embedding layer which projects it to a real-value vector of size 64. Then this vector goes through a first event-based SSM at its full resolution. The embedding is projected through the $\bar{\mathbf{B}}$ matrix which is square here, meaning a cost of 64×64 SOPs/event. The state update is made through the diagonal matrix $\bar{\mathbf{A}}$ thus negligible in SOP but the state is then projected further through the square matrix $\bar{\mathbf{C}}$ for again 64×64 SOPs/event. Considering the SSC dataset presents an average number of 8000 events per sample, as reported by Schöne et al. [34], the average SOP cost per sample for this first SSM block is already of $SOP_{first\ block} = 8000 \times (64 \times 64 + 64 \times 64) = 65.5M$.

The model then uses a pooling operation which divides the number of events going further in the network by 8, meaning 1000 events per sample in the second block. This block consists of two dense projections of 64×64 SOP/event and 3 more SSMs meaning $6 \times 64 \times 64$ SOP/event before the next event pooling. Altogether $SOP_{second\ block} = 1000 \times (2 \times 64 \times 64 + 6 \times 64 \times 64) = 32.8M$. The final pooling brings down the number of events to an average of 125 per sample for the final block, making it negligible in terms of SOP.

Altogether, the 64 state Event-SSM model requires $SOP_{total} \leq SOP_{first\ block} + SOP_{second\ block}$ which corresponds to 98.3M SOPs/sample. One can simply adapt this computation for the 128 states SHD version of the Event-SSM model, leading to an even higher 288.8M SOPs/sample. These numbers are around 10 times higher than the measured values for our SNN models (Figure 3). Additionally, we note that the Event-SSM relies on full precision real-value number for all its activations, meaning that each synaptic operation requires a full precision multiply-and-accumulate (MAC) operation, whereas our binary spiking models can handle simple accumulate (AC) operations as synaptic values

are either summed when there's a spike or else omitted. Altogether, it appears that the Event-SSM is significantly more computationally expensive to run, especially on event-based neuromorphic hardware.

A.5 Performance scalability with network width and depth

We demonstrate another strong capacity of our C-SiLIF model, its scalability in depth and width. We evaluate the C-SiLIF accuracy on the SHD task with respect to number of layers and hidden units along with the AdLIF, cAdLIF and Resonate and Fire (RF) models. Reported accuracies are obtained from a 5 seeds grid search on learning rate, dropout and scheduler for each scaling configuration, and results are shown in Fig. 5.

C-SiLIF is the prevailing model with the best mean accuracy of 95.64% obtained for 3 layers and 1024 hidden neurons. Furthermore with 95.06% obtained for 2 hidden layers of 128 neurons, C-SiLIF establishes a new state of the art performance on SHD for models below 50k parameters. Our model is only surpassed by the cAdLIF neuron for 5 hidden layers and displays an advantageous tendency in scalability both in terms of depth and width, corroborating the hypothesized beneficial stability property of the SSM-inspired features. We also observe that with some hyperparameter tuning, the AdLIF model appears better than reported in [7] and [21] but its poor scaling with depth confirms the claims on its lack of robustness.

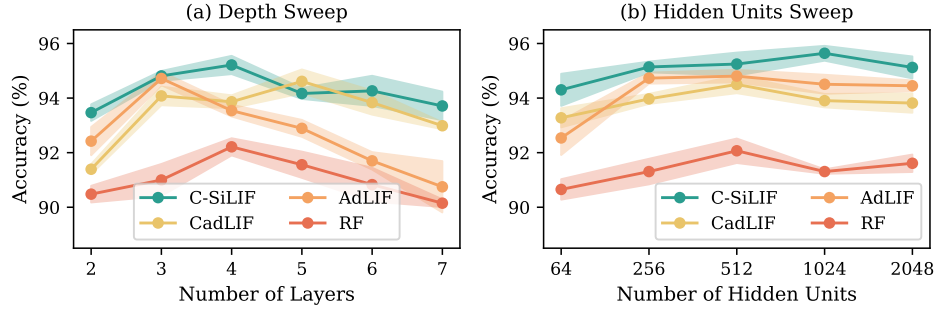


Figure 5: SHD test accuracy with standard deviation interval on 5 runs for models over depth with 128 fixed hidden units (*left*) and width with 3 fixed layers (*right*).

A.6 Incremental C-SiLIF features analysis

We carry an extra study of our C-SiLIF model, evaluating its performance with incrementally more SSM inspired features, starting from the Resonate and Fire (RF) neuron all the way to the C-SiLIF. We use networks of 2 hidden layers of 512 units and perform a grid search for each intermediate version. The measured average and standard deviation accuracy on the SHD dataset are reported in Figure 6.

As described by Izhikevich [23], the RF neuron follows these neuronal dynamics:

$$\begin{aligned} u_t &= u_{t-1} + \Delta t((\alpha^{real} + i\alpha^{img})u_{t-1} + I_t) - \theta s_{t-1} \\ s_t &= (Re(u_t) \geq \theta), \end{aligned} \quad (7)$$

where α^{real} and α^{img} are the trainable parameters, obtained directly from the discrete form. As for the gap between the AdLIF and our SiLIF model, a first difference with the C-SiLIF is the parametrization of the model, as the C-SiLIF model focus on training continuous state transition parameters. The first incremental feature on top of the base RF model, called *Continual domain* is thus to train λ^{real} and λ^{img} which are then converted back to the discrete transition variables $\alpha = exp((- \lambda^{real} + i\lambda^{img})\Delta t)$ such that $\alpha^{real} = Re(\alpha)$ and $\alpha^{img} = Im(\alpha)$. We note that this step slightly improves the model's performance.

The next step consists of making the timestep Δt a trainable parameter. Despite this feature having been identified as a central element of the success of our models (see ablation results in Table 3), this incremental step hinders the model's performance. But as soon as the logarithmic reparametrization is

included again, where λ^{real} and Δ_t are trained in the logarithmic domain, the model outperforms the previous *Continual domain* configuration. We thus hypothesize that these two features work in a strong synergy and should be considered as one common method.

Further on, the model’s parameter initialization is switched from the standard uniform distribution and range defined by Higuchi et al. [26] in their modern study of the RF model, to the *S4D-Lin initialization* from the S4 model [9]. This leads to a significant jump with the accuracy reaching a level above our own optimized reproduction of the AdLIF model. This reinforces the idea that the specific S4 initialization has been optimized effectively for this specific model parametrization, and thus serves our C-SiLIF model.

Finally our proposed half reset and input gating b (see Algorithm 2) complete the C-SiLIF model and improve the final performance further. This all demonstrates the positive impact of the S4 inspired features along with their strong synergy for SNN models.

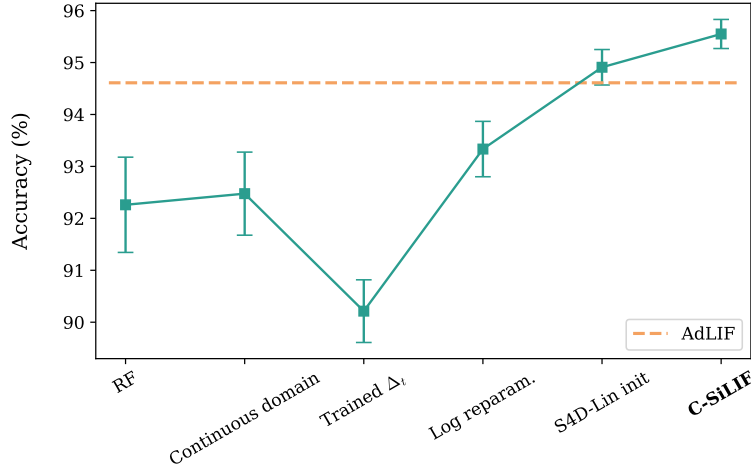


Figure 6: Impact of incremental SSM-imported features on performance on the SHD dataset from the Resonate and Fire (RF) to our proposed C-SiLIF model. Each model’s performance is obtained after a grid hyperparameter sweep for 2 hidden layers of 512 neurons. Errorbars correspond to the standard deviation on 5 seed runs.

A.7 Sparsity analysis

On top of the synaptic operations shown in Figure 3, we report here the specific sparsity numbers for different SNN models in Table 5 for reference. The sparsity correspond to the average amount of non-spiking timesteps per neuron produced with respect to the amount of total timesteps per sample.

Table 5: Test sparsity at best validation epoch for main models on SHD and GSC datasets, all with 2 hidden layers of 512 units. Standard deviation is reported with $\pm$.

Dataset	AdLIF	cAdLIF	C-SiLIF	SiLIF
SHD , 4ms resolution	88.57 $\pm$ 0.56%	97.80 $\pm$ 0.13%	80.25 $\pm$ 0.33%	98.66 $\pm$ 0.09%
SSC , 4ms resolution	N/A	93.65 $\pm$ 0.11%	90.37 $\pm$ 0.28%	93.35 $\pm$ 0.20%
GSC , 10ms resolution	93.39 $\pm$ 0.65%	93.03 $\pm$ 0.02%	85.21 $\pm$ 0.47%	91.71 $\pm$ 0.20%